



CS361: Assignment 2: Environment Setup, Course Project Plan, and Sprint 1 Plan (for Milestone #1)

Overview

Now that you've been introduced to the microservices concept, start planning your course project (except for the microservice you'll later make for a teammate) and your Main Program (Milestone #1). It's OK to change your plan later!

This assignment has three parts:

- **Part 1:** Environment Setup. Initialize your GitHub repository and investigate task management systems to organize your project tasks.
- **Part 2:** Course Project Plan. Write all the user stories you would like to be part of your course project (except for the microservice you'll later make for a teammate). It's OK if you don't implement all of them this term.
- **Part 3:** Sprint 1 Plan. Select at least three user stories to implement during Sprint 1 (for your Main Program / Milestone #1). Define detailed requirements for each user story.

Note these minimum requirements for Milestone #1:

- Your Main Program implementation must offer value to users
- At least three user stories are completed
- All features that are part of the milestone must be working. The milestone must not have partially completed features.
- It allows users to interact (e.g., provide input, push buttons, etc.)
- Reflects each of the Inclusivity Heuristics
- Reflects three quality attributes of your choice (i.e., satisfies the non-functional requirements you write for each quality attribute)

Hint: If you choose "usability" or "inclusivity" as a quality attribute, your corresponding non-functional requirement can involve the Inclusivity Heuristics.

Part 1: Environment Setup

Set up your development environment. In addition to an IDE or code editor (choose any you prefer), start a GitHub repository and choose a task management system.

Complete each item below by replacing the **highlighted text** (**Usability note:** double-click the text to select it).

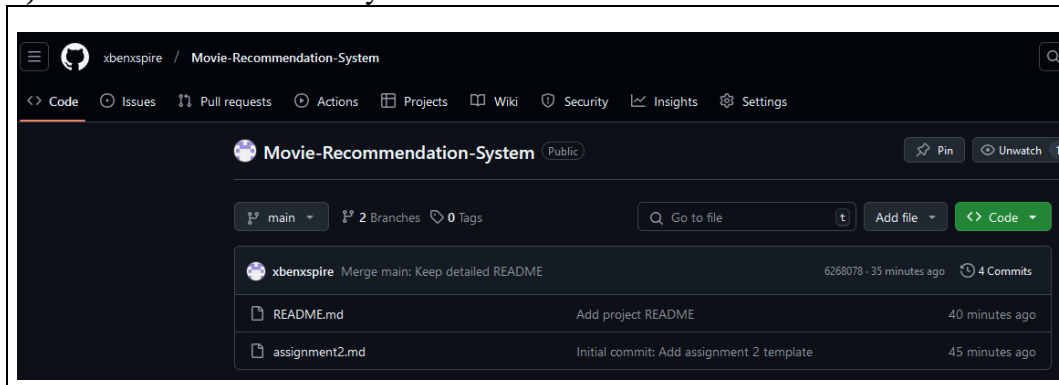
1) GitHub Repository

Create a GitHub account if you don't already have one, create a Git repository hosted on GitHub. This first repository will be for your Main Program. Make a **test commit**. The test commit should show up on GitHub.

a) What is your GitHub username?

xbenxspire

b) Provide a screenshot of your test commit.



Later, it would make sense for you to set up additional repositories, one for each of the microservices you'll implement.

2) Spike: Task Management Systems

For your course project, you will be using a task management system to keep track of development tasks. Spike at least **three** task management systems you could use.

A spike is a quick, directed effort focused on getting a question answered. Performing a spike can help you make intelligent decisions. Spikes do take time upfront, but they can also save you from making a bad choice that takes much more time to recover from. This portion of the assignment provides an opportunity to do a spike while making a relatively low-stakes decision (which task management system to use).

Examples of task management systems you could spike: Trello, Jira, Asana.

Requirements for the task management systems:

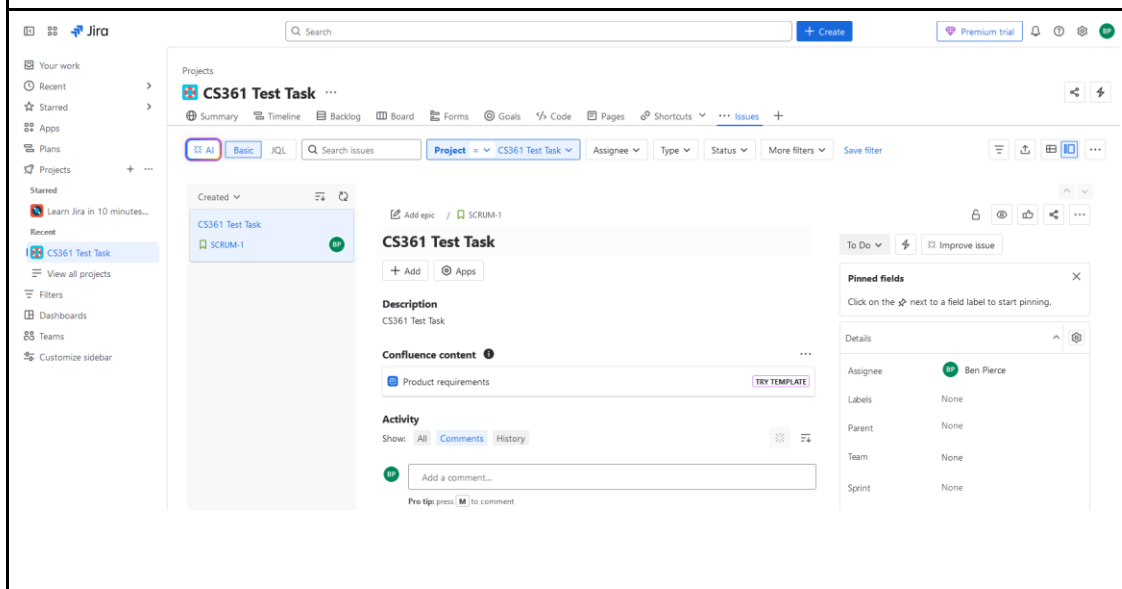
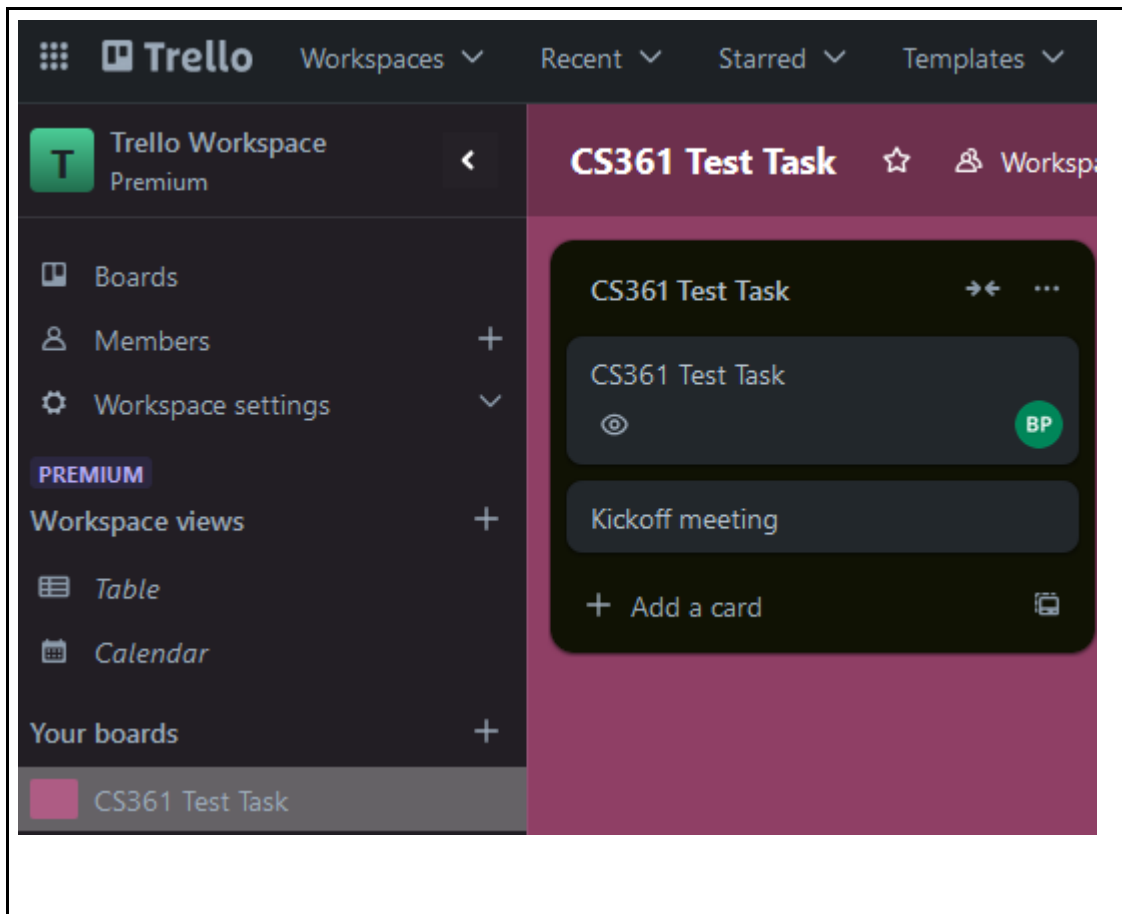
- Software specifically designed for task management
- Support for collaboration, task definition/deletion/updating, task priorities, task due dates, assigning people to tasks, setting task status, and the ability to label/tag tasks or put them in different columns.

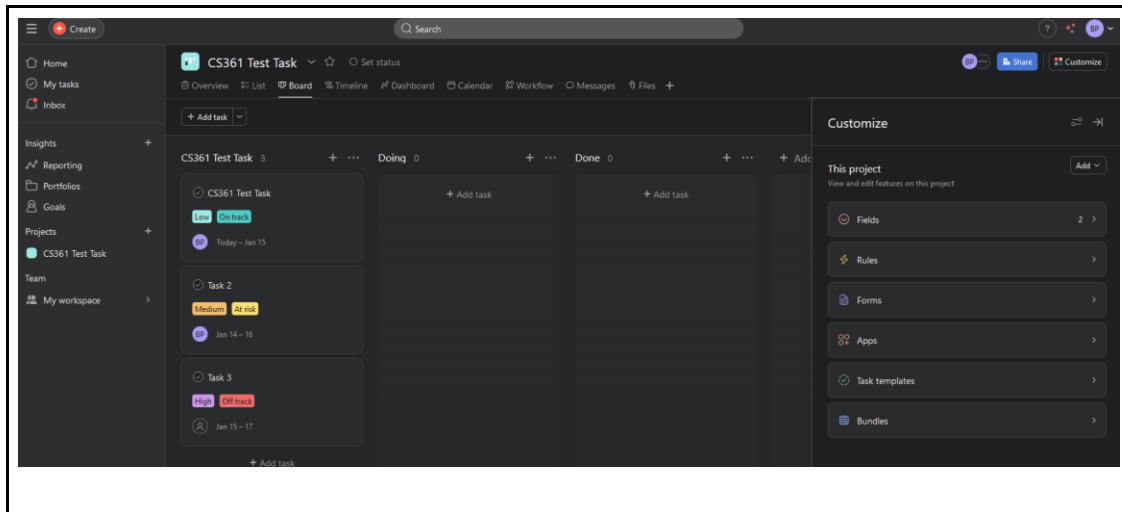
To do a spike, you need to research the task management systems and also (1) try to **use** them, (2) **evaluate** them based on specific criteria, (3) **compare** them, and (4) **decide** which to use.

a) Which task management systems did you spike?

<i>Trello</i>
<i>Jira</i>
<i>Asana</i>

b) **Try** each system. Create a task then update it, assign it, delete it, etc. **Screenshot** your task in each system and paste it below. Name the tasks "**CS361 Test Task**".





c) For each, **evaluate** against at least these criteria:

Ease of use. Ex: Is it intuitive to learn? Easy to remember how to use it? Do you find yourself making lots of errors trying to use it? Are there tutorials and documentation?

Name of system 1: *Trello*

Evaluation of system 1's ease of use (2+ sentences): *Trello offers a drag-and-drop interface that's understandable. The Kanban board layout makes it easy to visualize task flow, and the learning curve is minimal with helpful tooltips and guides.*

Name of system 2: *Jira*

Evaluation of system 2's ease of use (2+ sentences): *Evaluation of system 2's ease of use: Jira has a steeper learning curve with its extensive feature set offering comprehensive project management tool. However, the interface can feel overwhelming and requires time to learn all the available features.*

Name of system 3: *Asana*

Evaluation of system 3's ease of use (2+ sentences): *Evaluation of system 3's ease of use: Asana provides a clean, modern interface that balances functionality with simplicity. It offers multiple view options (list, board, timeline) and includes helpful onboarding tutorials.*

Speed/responsiveness. Ex: Does it take an annoyingly long time to log in / load / create new projects / etc. or is it peppy?

Name of system 1: *Trello*

Evaluation of system 1's speed/responsiveness (2+ sentences): *Trello loads*

quickly and updates in real-time. Card movements and updates are instant, and the interface remains responsive even with multiple boards and cards.

Name of system 2: *Jira*

Evaluation of system 2's speed/responsiveness (2+ sentences): *Jira can be slower to load initially and when switching between different views. The robust feature set sometimes impacts performance, especially when dealing with large projects.*

Name of system 3: *Asana*

Evaluation of system 3's speed/responsiveness (2+ sentences): *Asana maintains good performance with quick loading times. The interface updates smoothly, though some features like timeline views can take longer to load with many tasks.*

Feature set. Ex: Besides the required features, does the system have other features you are likely to need?

Name of system 1: *Trello*

Evaluation of system 1 (2+ sentences): *Trello offers essential task management features with a focus on simplicity. The Power-Up system allows adding specific features as needed, though some advanced project management features require paid plans.*

Name of system 2: *Jira*

Evaluation of system 2 (2+ sentences): *Jira provides the most comprehensive feature set, including advanced agile tools, detailed reporting, and customization options. It's particularly strong for software development projects.*

Name of system 3: *Asana*

Evaluation of system 3 (2+ sentences): *Asana balances features well, offering task dependencies, custom fields, and multiple project views. It includes team collaboration features and workflow automation tools.*

Relevance/popularity. Ex: Is it likely you will ever see the task management system again after the course?

Name of system 1: *Trello*

Evaluation of system 1's relevance popularity (2+ sentences): *Yes, Trello is widely used across various industries and is particularly popular in startups and small teams. Its simplicity makes it relevant for both personal and professional use.*

Name of system 2: *Jira*

Evaluation of system 2's relevance popularity (2+ sentences): *Yes, Jira is the industry standard for software development teams and is used by many major tech companies. Experience with Jira is highly valuable for software engineering careers.*

Name of system 3: *Asana*

Evaluation of system 3's relevance popularity (2+ sentences): *Maybe, Asana has growing adoption in professional settings, especially among marketing and creative teams. It's becoming increasingly relevant in tech companies for non-engineering teams.*

- d) **Compare** the systems by **ranking** them based on the criteria above. Best to worst for each criterion. **List or table format.**

System 1 name: *Trello*

System 1 ease of use: *1*

System 1 speed/responsiveness: *1*

System 1 feature set: *3*

System 1 relevance/popularity: *2*

System 2 name: *Jira*

System 2 ease of use: *3*

System 2 speed/responsiveness: *3*

System 2 feature set: *1*

System 2 relevance/popularity: *1*

System 3 name: *Asana*

System 3 ease of use: *2*

System 3 speed/responsiveness: *2*

System 3 feature set: *2*

System 3 relevance/popularity: *3*

- e) Which system is the **highest ranked**?

<i>Trello</i>

Decide which task management system you're going to use and use it to complete Parts 2 and 3.

Answer: Trello

Part 2: Course Project Plan

Write an initial set of user stories for your course project (except for the microservice you'll later make for a teammate). Put the user stories in a **Product Backlog** column/section/category of your task management system, or a label/tag the user stories "Product Backlog". You might end up changing these user stories later or adding new ones.

Complete each item below by replacing the **highlighted text** (**Usability note:** double-click the text to select it).

1) Product Goal and Backlog

You'll be using *some* Scrum methods in this course. Unfortunately, the Scrum Master and Product Owner roles don't work well in this course setting. You will, however, experience Scrum Events and Artifacts.

- a) What is your **Product Goal** for your course project? This includes your Main Program, Microservice A that your teammate will later implement for you, and Microservices B, C, and D. It does NOT include the Microservice A you will make for a teammate.

Develop a comprehensive movie recommendation system that helps users discover personalized movie suggestions through an intuitive interface, leveraging microservices architecture to provide real-time recommendations based on user preferences, viewing history, and genre analysis.

The Scrum Guide (<https://scrumguides.org/scrum-guide.html>) doesn't give a detailed description of the Product Goal: "**describes a future state**", "**long-term objective**".

Example Product Goal: "Develop a desktop app that listens to what people are saying and automatically shows content that might be relevant to their conversation."

- b) **Write user stories** for your **entire course project** (except for the microservice you will make for your teammate). Use INVEST to guide you.

Assignment requirements for Product Backlog user stories:

- Each has a **name** that briefly describes the functionality (e.g., "Login")
 - Each uses the "**As a... I want to... so that...**" format (explained in textbook)
 - Each is about **functionality** and not about the quality of the functionality or a constraint (user stories are functional requirements, not non-functional requirements)
 - Total of at least **10** user stories
 - As a set, must have **no obvious violations of INVEST**
-
- **User story 1:** *Search movies*
 - **User story 1:** As a movie enthusiast, I want to search for movies by title so that I can quickly find specific films I'm interested in.

- **User story 2:** *View Movie Details*
- **User story 2:** As a user, I want to view detailed information about a movie so that I can learn more about its plot, cast, and ratings.

- **User story 3:** *Rate Movies*
- **User story 3:** As a user, I want to rate movies I've watched so that I can keep track of my opinions and improve future recommendations.

- **User story 4:** *Get Recommendations*
- **User story 4:** As a user, I want to receive personalized movie recommendations so that I can discover new films that match my interests.

- **User story 5:** *Create Wishlist*
- **User story 5:** As a user, I want to save movies to a watchlist so that I can keep track of films I want to watch later.

- **User story 6:** *Filter Recommendations*
- **User story 6:** As a user, I want to filter movie recommendations by genre so that I can find specific types of movies I'm in the mood for.

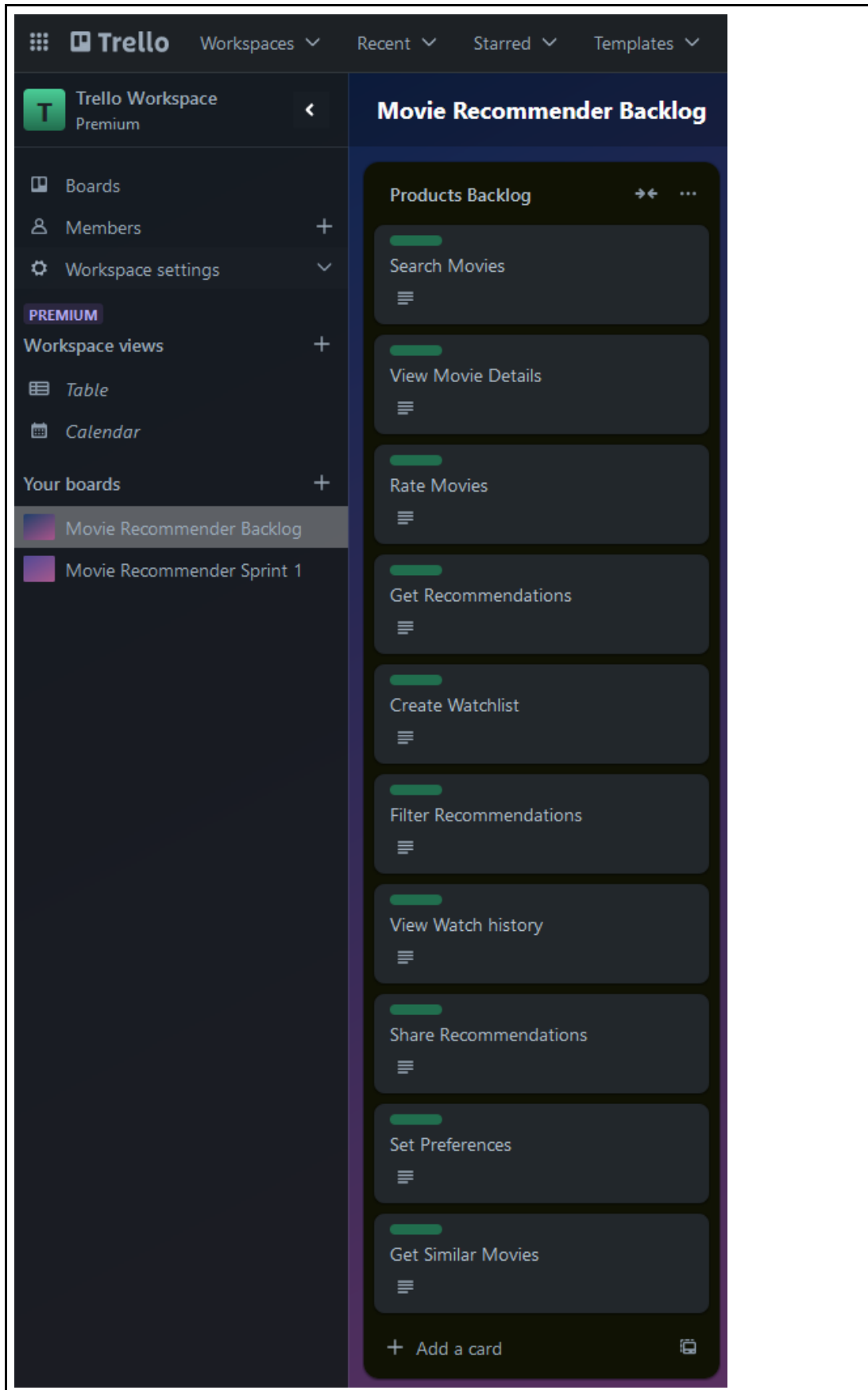
- **User story 7:** *View Watch History*
- **User story 7:** As a user, I want to view my watch history so that I can remember what movies I've already seen.

- **User story 8:** *Share Recommendations*
- **User story 8:** As a user, I want to share movie recommendations with friends so that we can discuss films we might enjoy together.

- **User story 9:** *Set Preferences*
- **User story 9:** As a user, I want to set my genre preferences so that I receive more relevant movie recommendations.

- **User story 10:** *Get Similar Movies*
- **User story 10:** As a user, I want to see similar movies to ones I've enjoyed so that I can find more content I might like.

Enter the user stories into your task management system in a **Product Backlog column/section/category**, or with a “Product Backlog” label/tag. Paste a **screenshot** below so that the grader can confirm you added the stories.



2) Quality Attributes

Quality attributes can help guide the entire development of your project. They can remind you (and other developers) what aspects of your project matter the most and can help you decide which features to implement and in what way.

Select the top three quality attributes you care about for your course project. See https://en.wikipedia.org/wiki/List_of_system_quality_attributes for ideas.

a) **Which three quality attributes did you choose? Name and define each.**

- **Quality attribute 1:** *Usability*
- **Quality attribute 1 definition:** *The degree to which the system can be easily used by its intended users to achieve their goals effectively and efficiently.*

- **Quality attribute 2:** *Performance*
- **Quality attribute 2 definition:** *The speed and efficiency with which the system responds to user actions and processes data.*

- **Quality attribute 3:** *Reliability*
- **Quality attribute 3 definition:** *The ability of the system to perform its required functions consistently and accurately under normal conditions.*

b) **Why did you choose these quality attributes?** Explain how each quality attribute is particularly relevant to your project (1+ sentence per quality attribute)

- **Why quality attribute 1 is relevant to your project:** *Usability is crucial for this project as users need to easily search for movies and understand recommendations without confusion or frustration. The interface should be intuitive for users of varying skill levels.*
- **Why quality attribute 2 is relevant to your project:** *Performance is essential because users expect quick responses when searching for movies and receiving recommendations. Slow loading times or delayed updates would significantly impact user satisfaction.*
- **Why quality attribute 3 is relevant to your project:** *Reliability is vital because users depend on accurate movie recommendations and consistent system behavior. The system must maintain data integrity and provide dependable service to build user trust.*

Part 3: Sprint 1 Plan (for Milestone #1)

Next, move some user stories from your Product Backlog to your Sprint Backlog, or change the label/tag to “Sprint Backlog”. These will be the user stories you **WILL** implement during Sprint 1 (for Milestone #1 / your Main Program) and comprise your Sprint Plan. Your Milestone #1 Main Program implementation must offer value to users.

- 1) What is your **Sprint Goal**? The Sprint Goal must clearly communicate what you plan to work on (e.g. what pages, what functionality, etc)

Implement the core movie search and basic recommendation functionality, focusing on creating a usable and responsive user interface that allows users to search for movies, view details, and receive initial recommendations.

- 2) Select **at least three** user stories from your Product Backlog and move them to your Sprint Backlog, or re-label/tag them “Sprint Backlog”. Because you will be implementing these user stories during the Sprint, you need to write more specific requirements in the form of **acceptance criteria**.

Acceptance criteria can cover both functional and non-functional requirements. The non-functional requirements can serve to carry through your intention to reflect quality attributes.

Some developers write their user stories on 3” by 5” index cards: The user story name and “As a” format goes on the front of the card and the acceptance criteria can go on the back. **Example:**

(Front of index card)	
Automatic IMDB (“As a ... I want to ... so that ...”)	
As a user speaking during a conversation, I want to automatically see the IMDB.com webpage for the movie I’m talking about so that I can continue with my conversation and examine the webpage as needed.	
(Back of index card)	
Acceptance criteria	
Functional requirements (“Given... when... then...”)	
● Given a person is speaking in English at 60 dB or louder when the software is at least 80% sure it knows what movie the person is talking about, then it will open and focus the default web browser and navigate to the movie’s IMDB.com webpage.	
Quality attributes & Non-functional requirements	
● Responsiveness: Once the software is 80% sure about what movie is being spoken about, it will display the movie’s IMDB.com webpage within 3 seconds.	

Use this format to fill out each of your Sprint Backlog user stories.

Assignment requirements for Sprint Backlog user stories:

- For each of the three (or more) user stories...
 - The front of the card must contain the user story’s name and “As a... I want to... so

- that...” format
 - The back of the card must contain at least one functional requirement and each functional requirement must use the “Given... when... then...” format.
- Each of your three quality attributes must appear at least once on a user story’s “back of index card” and must be converted to a non-functional requirement.
- All of the functional and non-functional requirements must be testable.

Later, you will be asked to show that your functional and non-functional requirements are met.

First user story

(Front of index card)
Search Movies <i>As a movie enthusiast, I want to search for movies by title so that I can quickly find specific films I'm interested in.</i>
(Back of index card)
Acceptance criteria Functional requirements ● <i>Given the user is at the main menu when they enter a movie title in the search prompt, then a list of matching movies should be displayed.</i> Quality attributes & Non-functional requirements ● <i>Performance: Search results must be displayed within 2 seconds of submitting the search query.</i>

Second user story

(Front of index card)
View Movie Details <i>As a user, I want to view detailed information about a movie so that I can learn more about its plot, cast, and ratings.</i>
(Back of index card)
Acceptance criteria Functional requirements ● <i>Given a user has searched for movies when they select a movie number from the list then the detailed information for that movie should be displayed.</i> Quality attributes & Non-functional requirements ● <i>Usability: Movie details must be organized in clear sections with consistent formatting and readable text size.</i>

Third user story

(Front of index card)

Get Recommendations

As a user, I want to receive personalized movie recommendations so that I can discover new films that match my interests.

(Back of index card)

Acceptance criteria

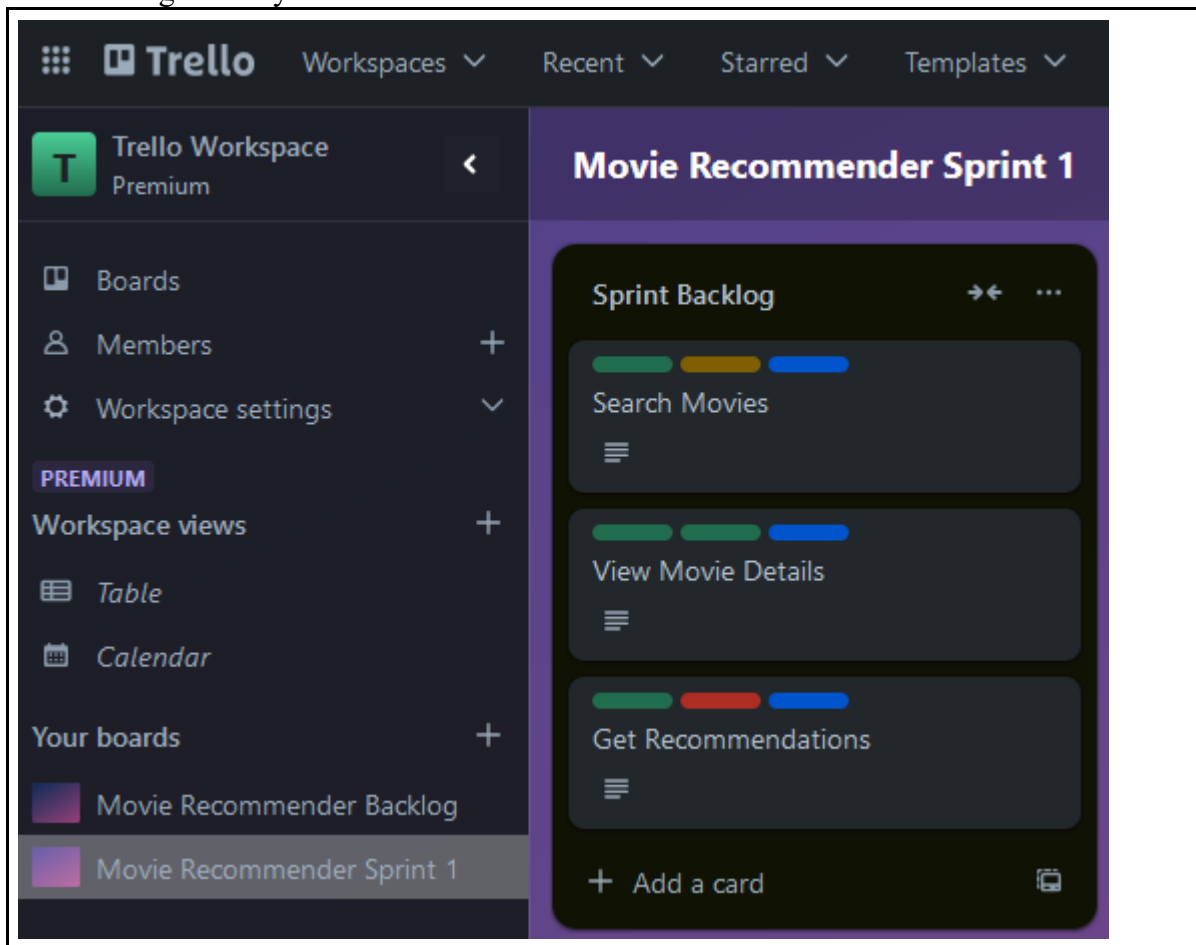
Functional requirements

- *Given a user has viewed at least one movie when they select the recommendations option from the main menu then a list of recommended movies based on their viewing history should be displayed.*

Quality attributes & Non-functional requirements

- *Reliability: The recommendation system must maintain 99% uptime during operating hours and provide consistent recommendations for the same user profile.*

- 3) Take a **screenshot** that shows you've moved these user stories into a Sprint Backlog in your task management system.



Your **Definition of Done** for Sprint would typically include, “The acceptance criteria are satisfied for all Sprint Backlog user stories.” You aren’t required to write your DoD or put it in your task management system.

This would also be **a good time to break each of your user stories into a list of specific tasks** you need to complete. Task management systems are, as you might imagine, a great place to do that!

Submission

PDF or Word format via Canvas.

Grading

You are responsible for satisfying all criteria listed in the Canvas rubric for this assignment.

Questions?

Please ask via Ed so that others can benefit from the answer.